

ML Algorithm used for the model- Logistic Regression

1. Algorithm Type & Objective

Logistic Regression is a supervised learning algorithm used for classification tasks.

In this project, it is used for multiclass classification (predicting ratings 1–5 from review text).

The main objective is to model the probability that an input belongs to a particular class, and then assign the class with the highest probability.

2. How the Algorithm Works (Mathematical Intuition)

-Logistic Regression models the relationship between input features (TF-IDF text vectors) and class labels (ratings) by learning a set of weights.

-Instead of directly predicting the class, it predicts the probability of each class using the softmax function for multiclass problems.

Softmax function for class j:

$$P(y = j|x) = \frac{e^{w_j \cdot x + b_j}}{\sum_{k=1}^k e^{w_k \cdot x + b_k}}$$

Where:

x = input feature vector (TF-IDF values)

w_j, b_j = weight and bias for class j

k = total number of classes (here 5)

The algorithm adjusts the weights w_j and biases b_j to minimize the cross-entropy loss:

$$LOSS = - \sum_{i=1}^n \sum_{j=1}^k y_{ij} \log(\hat{y}_{ij})$$

Where:

$y_{ij} = 1$ if sample i belongs to class j , else 0

$\hat{y}_{ij}$ = predicted probability for class j

3. Key Hyperparameters

- multi_class

'multinomial' → directly handles multiple classes using softmax (recommended).

'ovr' → one-vs-rest classification (less efficient for many classes).

- solver

Optimization algorithm. Common options:

'lbfgs' (good for multinomial problems, default for multiclass).

'saga' (supports large datasets, can be faster).

- max_iter

Maximum iterations for the solver. Increase if model does not converge.

Default: 100, but often set to 1000 for text data.

- C

Small C → Strong regularization → Helps prevent overfitting but may underfit the data. Large C →

Weak regularization → Fits training data more closely but may overfit.

The right C value helps the model generalize better to unseen data.

-Tuning using Parameter Grid:

C: [0.01, 0.1, 1, 10, 100]

penalty: l2 (default for multinomial Logistic Regression)

Method: Used GridSearchCV with 5-fold cross-validation to test each parameter combination.

Scoring Metric: Accuracy was used to compare model performance.

Computation: Used n_jobs=-1 to run in parallel on all CPU cores for faster processing.

4. Strengths & Limitations

-Strengths

Works very well with high-dimensional sparse data like TF-IDF vectors.

Interpretable: coefficients can show which words are most influential for each class.

Fast training and prediction even for large datasets.

Often a strong baseline model for text classification.

-Limitations

Assumes a linear relationship between features and log-odds of the outcome.

May not perform well when classes are not linearly separable.

Can struggle with very complex non-linear patterns in text.

5. When to Use / When Not to Use

-When to Use

As a baseline model for text classification.

When you want interpretability of model coefficients.

For high-dimensional sparse data like TF-IDF vectors.

-When Not to Use

When the relationship between features and labels is highly non-linear (deep learning models may work better).

When working with very small datasets and many classes — may overfit if not regularized properly.